Assignment 2: Detecting Emotions with ML Algorithms

Cross-Dataset Generalization of Facial Expressions

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1 Introduction and research questions

This report studies facial expression recognition with classical machine learning on two datasets, FER2013 and KDEF, and asks which challenges are specific to each dataset and how well knowledge transfers between them. The FER2013 baseline classifiers are covered in Task 1. Tasks 2 to 6 cover the KDEF study, cross-dataset generalization, the fuzzy classifier, the creative challenge and the real-time demo.

The primary question is how datasets and feature/model choices affect within-dataset accuracy and cross-dataset generalization. Secondary questions concern which emotions and representations are most transferable, whether interpretable facial geometry is useful, and how reliable a trained model remains under webcam conditions.

Declaration of AI tool use

Technology Statement: During the preparation of this work, we used OpenAI ChatGPT (GPT-5.6 Sol), Google Gemini (3.1 Pro), and Claude models (Sonnet 5 Medium and Haiku 5.5 Medium), in order to brainstorm assignment direction, support on coding and discuss specific technical concepts to support the team along the development. The following parts of the assignment were affected/generated by AI tool usage: TASK 1 (section 3) / TASK 2 (section 4) / TASK 3 (section 5)/ TASK 4 (section 6) / TASK 5 (section 7) / TASK 6 (section 8). After using this tool/service, Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto take full responsibility for the content of their work.

2 Datasets, preprocessing and experimental protocol

FER2013 provenance and protocol: FER2013 was introduced by Goodfellow et al. (2013) as part of a Kaggle challenge. Images were collected automatically from web image searches and labelled with seven emotion categories, and the authors report human accuracy of roughly 65%. We used the provided Training, PublicTest and PrivateTest splits without re-splitting. The Training split fitted all models, PublicTest was used for model comparison and tuning, and PrivateTest was used once per tuned model. All classifiers use seed 42, and the bootstrap intervals use 1,000 resamples (seed 0). The software versions used for each task are listed in Table 8 (Appendix B.1).

Preprocessing and face detection: FER2013 images are already cropped 48×48 grayscale faces, so no detection or resizing was applied. HOG was computed on the raw image, and PCA used pixels scaled to $[0, 1]$. For landmarks, the whole image is passed to *dlib* as the face box, so there are no detection failures by construction. The cost is that landmark quality depends on the crop being well aligned, which fails for tilted, occluded or off-centre faces (Figure 8 in Appendix C).

Quality checks: Exact duplicates were found by MD5 hashing. 280 of 3,589 validation images (7.8%) and 288 of 3,589 test images (8.0%) also appear in the training split. We kept duplicates in all Task 1 evaluation splits, so results stay comparable to the provided protocol, and we report non-duplicated scores separately (Section 3). Duplicates were removed from the training set only in Task 5 (section 7).

Limitations: The dataset has strong class imbalance (Disgust is 1.5% of training, see Figure 6 and Table 10 in Appendices C and B), noisy or ambiguous labels, low resolution, and cross-split leakage through duplicates.

Examples of the first two issues are shown in Figure 8 (Appendix C).

KDEF. This database was presented by Goeleven et al. (2008), but for this assignment, we use only straight/frontal (S) faces, across seven comparable expressions (KDEF "Afraid" maps to FER "fear"). Divide by identity: 80% of subjects for training, 20% for testing, with no subject overlap.

Common representation: Detect and crop KDEF faces with a margin preserving forehead and jaw; convert to grayscale, resize to 48×48 and apply the same normalization as FER. Apply identical feature extraction at training and inference. Fit any scaling, PCA or learned transformation only on training data.

HOG features: Histogram of Oriented Gradients (HOG) describes the local edge directions of an image. Each 48×48 image is described by 900 HOG features (9 directions in each of 2×2 cells, over 5×5 overlapping block positions). The derivation is given in Appendix B.2.

Fuzzy classifier protocol: For Task 4, geometric features were extracted directly from original-resolution frontal KDEF images, rather than the FER-compatible 48×48 images. Both acquisition sessions for each identity remained in the same partition. Model selection used training subjects only, preserving the held-out KDEF subjects for final evaluation.

3 Task 1: FER2013 baseline

Features. Three feature types were extracted, each compared with three classifiers ($3 \times 3 = 9$ models):

- **HOG (900 dimensions):** histogram of oriented gradients on the 48×48 image, 9 orientations, 8×8 pixel cells, 2×2 blocks with L2-Hys normalization. HOG encodes local edge orientation (mouth corners, brows, eye shape, wrinkles) and the block normalization gives some robustness to illumination.
- **Facial landmarks (136 dimensions):** the 68 dlib landmarks, centred and scaled to $[-1, 1]$. The whole 48×48 image is passed to dlib as the face box, so the same procedure can be reused in the real-time demo.
- **PCA (100 dimensions):** raw pixels scaled to $[0, 1]$, projected with whitened PCA. PCA is fitted inside the scikit-learn pipeline on the training data only, which avoids leakage into the validation and test sets.

Classifiers. Multinomial Logistic Regression ($C = 1$, standardized inputs), an RBF-kernel SVM ($C = 1$, $\gamma =$ "scale", standardized inputs) and a Random Forest (300 trees, sqrt features, no scaling). All use a fixed random seed (42).

Results on the validation set. Table 9 in Appendix B shows accuracy, balanced accuracy and macro-F1 for all nine combinations. Macro-F1 was the selection criterion, because it weights all seven classes equally and is therefore not dominated by Happy.

Best classifier. The RBF-SVM is the best classifier on two of the three feature types (HOG: 0.505 macro-F1; PCA: 0.430) and the best overall. Its non-linear kernel can model the non-linear relation between local gradient patterns and expression, which the linear Logistic Regression cannot: Logistic Regression is the weakest or near-weakest classifier for every feature type (0.287 to 0.395 macro-F1). The Random Forest is competitive only on landmarks, where it is the best classifier (0.474) and clearly beats the SVM (0.401). A plausible but untested reason for the landmark result is given in Appendix B.11.The SVM is also by far the most expensive model to train (438 s on HOG, against 36 s for the HOG forest and 23 s for Logistic Regression).

Best features. HOG is the best representation for Logistic Regression and the SVM (for example 0.505 against 0.430 for PCA and 0.401 for landmarks with the SVM). For the Random Forest, landmarks are the best representation (0.474 against 0.440 for HOG). We attribute the HOG result to its local edge information, and the reasons for each ranking, including the sensitivity of landmarks and PCA to misalignment, are discussed in Appendix B.11.

Hyperparameter tuning and final test result. Each classifier was tuned on the feature set where it did best on validation, and tuning raised the SVM from 0.505 to 0.562 validation macro-F1. The search grids and the tuned validation scores for every classifier are given in Appendix B.10. Each tuned model was then evaluated once on the private test split (Table 1).

Table 1: Final test performance of each tuned classifier on the PrivateTest split, with 95% bootstrap intervals for macro-F1 (1,000 resamples). The confusion matrices are in Appendix A.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1	95% interval
HOG	SVM ($C = 10$)	0.575	0.550	0.568	[0.544, 0.590]
Landmarks	Random Forest	0.513	0.463	0.482	[0.457, 0.507]
HOG	Logistic Regression	0.438	0.376	0.378	[0.355, 0.400]

The test score of the SVM is higher than its validation score (0.568 against 0.562 tuned, and 0.505 default).

The paired bootstrap difference between the SVM and the landmark Random Forest is +0.086 with interval $[+0.062, +0.111]$, so the SVM's lead on the test split is unlikely to be noise. The gap between accuracy and balanced accuracy on validation (0.542 against 0.482) is caused mainly by the low recall of Disgust (25%).

Which emotions are hardest to distinguish? Averaging per-class F1 over the nine validation models gives the following ordering from hardest to easiest: Fear (0.26), Disgust (0.27), Angry (0.32), Sad (0.38), Neutral (0.43), Surprise (0.62), Happy (0.64). For the SVM on the test set, F1 is 0.75 for Happy and 0.73 for Surprise, and 0.43 to 0.46 for Angry, Sad and Fear.

- Happy and Surprise are relatively easier to classify. Both have distinctive, large-scale configurations: a raised, widened mouth with visible teeth for Happy, and wide-open eyes and mouth for Surprise. Happy additionally has the most training data.
- Fear, Angry and Sad are harder and are confused with each other and with Neutral. The most frequent errors of the validation SVM are listed in Appendix B.11; the test set shows the same pattern (Appendix A). These classes are expressed through subtle, low-intensity changes around the brows and eyes, which are hard to resolve at 48×48 pixels. Fear also shares features with Surprise (wide eyes, open mouth), and Sad and Neutral faces are visually close, so Neutral is a frequent "sink" class.
- Disgust is rare (436 training images), so the classifier rarely predicts it. On validation the SVM has precision 1.00 but recall 0.25: when it predicts Disgust it is right, but it misses most Disgust faces. On the test set recall is 0.51, but with only 55 test images this estimate is noisy. Class balancing was tested in Task 5 and did not help.

Evidence from error analysis. Illumination and contrast have a small effect on error, around five percentage points: the SVM error rate is about 0.46 to 0.49 in the three darkest brightness quintiles and about 0.42 to 0.44 in the two brightest (Figure 7, Appendix C). Recurring error types, including ambiguous or questionable labels, occlusion, non-frontal pose and overlaid text or watermarks, are illustrated in Figure 8 (Appendix C). Both figures are qualitative evidence rather than counts.

Duplicates. FER2013 contains exact copies of images across splits, and some copies carry different labels, which is direct evidence of label noise. Duplicates inflate the test score: on the 3,301 non-duplicated test images the SVM reaches 0.505 macro-F1, against 0.568 on the full test split, so 0.505 is the fairer estimate. The duplicate breakdown and the validation effect of removing duplicates are given in Appendix B.21. We treat cleaning the training set as a data curation step in Task 5.

Summary and limitations. The tuned HOG + RBF-SVM is the best FER2013 baseline (test: 0.575 accuracy, 0.550 balanced accuracy, 0.568 macro-F1; 0.505 macro-F1 without duplicates). HOG is the strongest feature type for Logistic Regression and the SVM, and landmarks are the strongest for the Random Forest; the SVM is the strongest classifier on HOG and PCA. Happy and Surprise are recognized reliably, while Fear, Angry, Sad and Disgust are not. Prediction times for the demo are given in Appendix B.11.Limitations: (i) the SVM was tuned on a 10,000-image subsample; (ii) the final test was run once per model, with no further selection; (iii) no class balancing was used, which disadvantages Disgust; (iv) the landmark features depend on a detector applied to 48×48 images with the whole image as the face box.

4 Task 2: KDEF dataset study

In this task we take the best FER2013 representation from Task 1, HOG, retrain the three classifiers on KDEF, and compare the results with FER2013.

The Karolinska Directed Emotional Faces (KDEF) dataset contains 4,900 pictures of 70 individuals displaying 7 emotions, photographed in a controlled studio setup. The pose of most interest to us is *straight*; the full collection protocol is described in Appendix B.23.

KDEF images (562×762 RGB) are converted to the FER2013 format (48×48 grayscale, scaled to $[0, 1]$) with the common representation of Section 2; the face-detection details are given in Appendix B.23.

The data is split 80/20 by subject i.e out of the 70 subjects, 56 of them are used in the training set, and 14 are in the test set. This is to ensure there is no data leakage into the test set. KDEF is perfectly balanced, with 112 training and 28 test images per emotion (Figure 10 in Appendix C).

In the previous task, we found out that the best feature representation is Histogram of Oriented Gradients (HOG), so every KDEF image is described with the same 900 HOG features (Section 2).

4.1 Model training and selection

We train the same three classifiers as in Task 1: Logistic Regression, SVM and Random Forest. For each classifier we run a grid search with 5-fold cross-validation, where the folds are also split by subject (using GroupKFold),

so the model is always validated on subjects it has not seen during training. The search space (20, 96 and 84 configurations) is listed in Table 11 in Appendix B; the selected settings are shown in bold in that table.

4.2 Results

	FER2013		KDEF	
Classifier	Validation	Test	Validation	Test
Logistic Regression	0.395	0.378	0.669	0.581
SVM	**0.505**	**0.568**	**0.679**	0.604
Random Forest	0.440	–	0.611	**0.614**

Table 2: Macro-F1 with HOG features, for the models trained and evaluated on FER2013 (Task 1) and on KDEF.

Table 2 shows that all three classifiers score higher on KDEF than on FER2013, even though they are trained on only 784 images instead of about 28,700. The reason is that KDEF is a much cleaner dataset. Every photo is taken in the same studio, with the same lighting and camera position, and the actors show each emotion clearly, so the faces line up well and the expression is almost the only thing that differs between images. FER2013 images come from the web, with different lighting, head poses, faces partly covered by hands or glasses, and labels that are sometimes wrong, so the classifiers have to learn the same emotion from very different-looking images.

4.3 Assignment Questions

Why are KDEF images easier or harder to classify? Overall, KDEF is easier: with the same HOG features, the SVM reaches a macro-F1 of 0.679 in cross-validation and 0.604 on the test subjects. The consistency of the KDEF images that makes this possible is described in Appendix B.12.

The mean face per emotion for both datasets (Figure 11, Appendix C) shows how consistently each facial feature is positioned across images: the blurrier the mean face, the less consistent the images for that emotion.

Which models change ranking? Table 3 compares the rankings.

Rank	FER2013 validation	KDEF cross-validation	KDEF test
1	SVM (0.505)	SVM (0.679)	Random Forest (0.614)
2	Random Forest (0.440)	Logistic Regression (0.669)	SVM (0.604)
3	Logistic Regression (0.395)	Random Forest (0.611)	Logistic Regression (0.581)

Table 3: FER2013 vs KDEF model comparison

On KDEF, Logistic Regression jumps from last to second place in validation, while the Random Forest falls to last. We think this is because the well-aligned KDEF faces make the HOG features almost linearly separable, so a strongly regularized linear model ($C = 0.01$) works well with few training images. Although, the Random Forest performs better in the test set by a small margin, we choose the best model from the validation set.

Which emotions transfer well between datasets? To see which emotions hold up on both datasets, we compare the F1 score per emotion of the KDEF models with that of the FER-trained SVM from Task 1 (Table 12 in Appendix B). Happy is the easiest emotion on both datasets (0.93 on KDEF and 0.75 on FER2013 for the SVM), and Surprise also scores well on both (0.62 and 0.73). Both expressions have large, distinctive shapes, a smile or an open mouth with raised eyebrows, which HOG picks up whether the expression is posed in a studio or caught in a web photo. Disgust scores well on both too, but FER2013 contains very few Disgust images (1.5%), so this result is less certain. Fear, Sad and Angry are among the hardest emotions on both datasets, with F1 scores between 0.33 and 0.55.

The confusion matrices in Appendix A (Figure 3) show that most mistakes happen between emotions that look alike. Fear is the clearest example: the SVM mistakes more fearful faces for Surprise (43%) than it gets right (36%), and the other two classifiers do the same. This is not surprising, since both expressions have wide-open eyes, raised eyebrows and an open mouth. In the same way, Sad is often mistaken for Neutral (21%), and Angry for Fear (25%) or Disgust (14%). Based on this, we expect Happy and Surprise to transfer best between the datasets, and Fear, Sad and Angry to transfer worst.

5 Task 3: Cross-dataset generalization

In this task, we will use the best model and feature extraction method and train on the entire KDEF dataset. We use this trained model to directly predict on the FER2013 dataset. The best model from Task 2 is HOG + SVM.

5.1 Results

Trained on	Tested on	Accuracy	Balanced accuracy	Macro-F1
KDEF	KDEF (Task 2, cross-validation)	0.686	0.686	0.679
FER2013	FER2013 (Task 1, PrivateTest)	0.575	0.550	0.568
KDEF	**FER2013 (full dataset)**	**0.225**	**0.215**	**0.195**

Table 4: Results of the HOG + SVM model for cross dataset

When we test the KDEF model on FER2013, performance of the model is worse (Table 4). Macro-F1 decreases from 0.679 to just 0.195, and balanced accuracy is 0.215. The chance level for seven classes is 0.14 (1 in 7), so the accuracy of 0.225 is only modestly above random guessing.

The confusion matrix (Figure 4 in Appendix A) shows where things go wrong. The model funnels most faces into just a few emotions, mainly Sad, Fear and Angry. For example, 39% of disgusted faces and 32% of neutral faces end up labelled as Sad. Neutral is hardly ever predicted, even though it's one of the most common emotions in FER2013. Only Happy and Surprise hold up somewhat better than the rest.

5.2 Assignment Questions

What adaptations did you have to make to run the model? As mentioned before, the format of images in both datasets is very different. The color map, dimensions etc. of KDEF are not the same as FER2013. As images of FER2013 are all grayscale, it cannot be converted into the same format of KDEF because we can't recover the missing colour information. Therefore, KDEF is converted it to grayscale, resized it to 48×48 pixels and scaled the pixel values to between 0 and 1. FER2013 images are already that size, so they only needed rescaling.

Once both datasets are pre-processed into the same format, we run the HOG feature extraction to convert the 48×48 into 900 features, which will be used by the SVM for classification.

What is domain shift? Domain shift is when a model is trained on one dataset and is tested on a different dataset. This helps in understanding how well the model has generalized the patterns for classification. This did not succeed well between the KDEF and FER2013 because of the following reasons:

- **Source of images** FER2013 images are publicly available images,and there is no selection criteria when the dataset was created. The only criteria was if the emotion is being displayed, and there is no consistency on the subject, the orientation, facial features/attributes. Emotion expression in FER2013 are less obvious, and the labels are not always correct. KDEF images were taken in a controlled environment, and all subjects were given explicit instructions during the data collection. The subjects in KDEF make each emotion easy to read.
- **Imbalanced dataset** KDEF has the same number of images for each emotion. FER2013 does not: 25% of its images are Happy, but only 1.5% are Disgust (Figure 12, left, in Appendix C). For a model like SVM, an imbalance in dataset can definitely affect its performance

We then checked whether these differences are also visible in the HOG features our model uses: a simple classifier trained to predict which dataset an image came from reaches a balanced accuracy of 98.8% (5-fold cross-validation), so the HOG features are different enough for a model to tell the datasets apart. The sampling and evaluation details are given in Appendix B.14.

Why does performance degrade? In short, the model was trained on images from controlled studio environment of 70 people facing the camera. FER2013 had no specific environment, and were images of random subjects

Our choice of features makes this harder still: HOG ties every feature to a fixed grid cell, so faces that are shifted, tilted or cropped differently (Figure 11, Appendix C) produce different gradient patterns for the same expression. The mechanism is detailed in Appendix B.15. This is also why a simple classifier could separate the two datasets with a balanced accuracy of 98.8% using HOG features alone. Examples of FER2013 images that the model gets wrong are shown in Figure 13 (Appendix C).

Some emotions are being predicted well (Table 13 in Appendix B). Similar to task 2, Happy has the best numbers (the reason is given in Appendix B.7). Disgust and Neutral have the worst numbers. The model does not almost recognize that emotion at all

Which features transfer best? Since our model relies only on HOG, we looked at which of its 900 features carry over best to FER2013. For each feature, we calculated its average value per emotion, once in KDEF and once in FER2013, and then compared the two patterns using a correlation. A correlation close to 1 means the feature rises and falls with the emotions in the same way in both datasets, so what the model learned from it still holds on FER2013.

Looking at Figure 14 and Table 14 (Appendix C and Appendix B), it is evident that the features around the centre of the face transfers better. Features around the eyes, mouth and cheeks, which are the regions that actually express emotion, keep almost the same pattern in both datasets, with correlations as high as 0.95. Most of the features with low correlation are outside the face.

If the central features transfer so well, why does the model still perform so poorly on FER2013? Correlation only captures part of the picture. It compares the average feature value for each emotion, but the model classifies individual images, not averages. Individual FER2013 images vary much more than KDEF images because of differences in alignment, pose and lighting, so even in the best features the emotions overlap heavily. The values themselves are also shifted relative to KDEF, which pushes many images to the wrong side of the boundaries the SVM learned. Finally, the SVM does not get to pick only the good features: it uses all 900 at once, including the edge features that point in the opposite direction.

6 Task 4: Explainable fuzzy classifier

Features and protocol. We extracted seven geometric features from the 68 dlib facial landmarks: mouth width, mouth opening, left and right eye opening, eyebrow height, mouth width-to-height ratio (*smile ratio*), and mouth-corner asymmetry. Most distances were normalized by face width, while eye opening was normalized by eye width. All 980 frontal KDEF images were processed successfully, without landmark detection failures.

The predefined subject-independent split (56 training and 14 held-out test subjects, with 44/12 fitting and validation subjects inside the training partition) is detailed in Appendix B.16. MinMax scaling was fitted only on the corresponding training partition.

Fuzzy model. We trained seven one-versus-rest Takagi-Sugeno fuzzy models with PyFUME, one for each emotion. PyFUME's default rule simplification and holdout were disabled; the altered parameters are listed in Appendix B.18.

Each model uses Gaussian membership functions in its rule antecedents and a learned linear consequent. The numerical outputs are combined through fuzzy inference, and the emotion with the highest output score is selected. These scores are not calibrated probabilities.

We compared two and three clusters per emotion using macro-F1 on validation subjects (Table 5). The three-cluster configuration was selected before testing and subsequently refitted on all 784 training images.

Table 5: Subject-independent KDEF fuzzy classification results.

Split	Clusters	Images	Acc. (%)	Bal. acc. (%)	Macro-F1
Validation	2	168	71.43	71.43	0.6881
Validation	3	168	73.21	73.21	0.7087
Final test	3	196	63.27	63.27	0.6082

Rule interpretation. The final system contains 21 fuzzy rules, three per emotion. A representative rule from the happiness model, given in full in Appendix B.9, requires all seven geometric measurements to belong to their first clusters, and then computes the happiness score from the fitted linear consequent.

This rule dominated 94.6% of happy training images but only 2.2% of other images. Its relatively wide-mouth configuration is consistent with smiling. The rules for fear and sadness separate less clearly (Appendix B.13).

Evaluation and limitations. On unseen subjects, happiness (28/28), surprise (26/28) and disgust (23/28) achieved the highest recall. Fear (7/28) and sadness (8/28) were more difficult: 14 fearful faces were classified as surprise, and 12 sad faces as neutral. The full confusion matrix is in Table 7 (Appendix A). The roughly ten-percentage-point accuracy decrease from validation to testing indicates limited generalization across individuals. Although the fuzzy antecedents are inspectable, the linear consequents and interactions between the seven models complicate explanations of individual predictions.

For comparison, Table 2 reports the Task 2 classifiers trained on HOG features and evaluated on the same 14 held-out KDEF subjects (196 images). The fuzzy model reaches a macro-F1 of 0.608, slightly above the SVM

selected on validation (0.604) and below the Random Forest (0.614), which scored best on the test subjects but was not selected on validation. The fuzzy model performs comparably while using seven interpretable measurements instead of 900 HOG features; its accuracy and the accuracy gaps to the other classifiers, which amount to a few images out of 196, are given in Appendix B.22. Its drop from validation (0.709) to test (0.608) cannot be compared directly with the SVM's (0.679 to 0.604) (Appendix B.22).

7 Task 5: Creative challenge

Starting from Task 1 baseline, each technique tested one change at a time, using a fast Logistic Regression and the validation split (PublicTest). Table 6 gives the ablation. The test split was used once, for the final model.
Changes tested.

- **Feature engineering (kept).** The 136 landmark values were appended to the 900 HOG features (1,036 features). HOG describes texture around the mouth and eyes, and landmarks describe the shape of the face, so the two carry different information.
- **Data cleaning (kept).** Training images were fingerprinted with MD5 hashes. Duplicates with the same label were reduced to one copy, and images whose copies disagreed in label were removed (107 images), reducing the training set from 28,709 to 27,434 images. The gain is within the noise level (Section 3).
- **Class balancing (rejected).** Balanced class weights lowered both macro-F1 and Disgust F1, so the switch was left off.
- **Augmentation (rejected).** Mirrored copies of the training images (with the landmarks reflected to match) doubled the training set. Macro-F1 went from 0.455 to 0.452 and Disgust F1 from 0.325 to 0.295, so the mirrored copies were left out of the final model.

Table 6: Validation (PublicTest) ablation with Logistic Regression. Rows 1 to 3 are cumulative. Class balancing (row 4) was tested on top of landmarks and cleaning, and was rejected. Mirrored augmentation (row 5) was tested on top of landmarks and cleaning, without class balancing, and was also rejected. The final model uses landmarks and cleaning only.

Variant	Accuracy	Macro-F1	Disgust F1	Training images
Baseline (HOG)	0.435	0.395	0.324	28,709
+ landmarks	0.497	0.453	0.325	28,709
+ cleaning (kept)	0.500	0.455	0.325	27,434
+ class balancing (rejected)	0.469	0.424	0.242	27,434
+ mirrored augmentation (rejected)	0.503	0.452	0.295	54,868

Final model. The tuned Task 1 SVM ($C = 10$) was retrained on the kept changes: 1,036 features and the cleaned training set. Because the feature count rose from 900 to 1,036, γ was rescaled by 900/1036 to keep the kernel width comparable ($\gamma \approx 9.7 \times 10^{-4}$). This rescaling is a heuristic and was not tuned. On validation, the model reaches 0.607 accuracy and 0.598 macro-F1, against 0.562 for the tuned Task 1 SVM. On the test split, evaluated once, it reaches 0.594 accuracy, 0.572 balanced accuracy and 0.592 macro-F1, against 0.568 for the Task 1 SVM. Its confusion matrix is in Figure 5 (Appendix A).

Interpretation. Adding landmarks is the main gain (+0.058 macro-F1 for Logistic Regression), and it is consistent with the landmark-based Random Forest in Task 1, which is the best classifier on landmarks. Cleaning changes the score by 0.002, which we treat as no effect. Class balancing hurt, even though Disgust recall was low (0.25) in Task 1. The notebook does not analyze why, so we leave the mechanism open. The test gain of 0.024 macro-F1 over Task 1 is not tested for significance, and the Task 1 bootstrap interval for macro-F1 is about ±0.023. Two limitations apply. The ablation uses Logistic Regression, not the final SVM, and each step is a single run. The test split still contains the 8% of images that also appear in training (Section 3). KDEF-derived features were not used in the final model.

8 Task 6: Real-time webcam demo

Our demo adapts the best selected model into a command-line executable Python file that opens the laptop webcam and performs live emotion recognition. We have chosen the model from **Task 4 - Fuzzy Classifier**, which was trained based on the *KDEF* database.

Run instructions for the demo are in the **README** file, and in Appendix B.19.

The seven features used in the live detection are identical to the ones reported in section 6. Detection, however, was done differently (Appendix B.19).

We performed an empirical evaluation where the following protocol was followed:

The protocol (refer to a KDEF reference, mimic the emotion, hold it for a while, and compute the results) is described in Appendix B.20.

The final experiment data can be visualised in Table 15 in Appendix C. The experiment summaries show that happy, sad, surprise, neutral, disgust and angry were classified in $> 80\%$ of the cases. Fear, on the other hand, was extremely difficult to trigger, and it was confused mostly with neutral and sad. Obviously, this experiment does not constitute a reliable evaluation metric for the live demo classifier. For that, it would be necessary to expose the model to data with emotions labelled properly.

Challenges and limitations: To improve throughput, our code downscales the whole frame to 640 px in width (for detection only). Landmarks are then taken from the full-resolution frame. This modification substantially increased the processing fram rate from 1.17 ($693.03ms$) to 13.71 processed FPS ($17.28ms$ average detection), running on MacOS.

In terms of luminosity, camera permissions and detection robustness, we did not experience any major issues, but mostly of the development for that section was done in MacOS. We foresee that other operating systems might add extra challenges during execution, which is a limitation the team could not address in this particular assignment.

9 Discussion, limitations and conclusion

FER2013. The tuned HOG + RBF-SVM was the best baseline (test macro-F1 0.568, or 0.505 without duplicated images). HOG was the best features for Logistic Regression and the SVM, and landmarks were best for the Random Forest. Happy (F1 0.75) and Surprise (0.73) are recognised reliably. Fear, Angry, Sad and Disgust are not (F1 0.43 to 0.46 for the first three), and they are mostly confused with Sad, Neutral and each other. Several errors trace to the data rather than the model: mislabelled or ambiguous images, occlusion, non-frontal pose, and duplicated images that inflate test scores. Lighting played a smaller role, with about a five-point change in error rate (Figure 7).

Creative challenge. Appending landmarks to HOG was the only change with a clear effect (+0.058 macro-F1 with Logistic Regression), and the final SVM reached 0.592 test macro-F1 against 0.568 for the baseline. The cost is more features (1,036 against 900), a heuristic γ rescaling that was not tuned, and extra training time. Duplicate cleaning had no measurable effect (+0.002). Class balancing lowered macro-F1 and Disgust F1, so Disgust remains weak. Disgust recall is 0.51 on a test class of only 55 images, so that figure is noisy. The ablation used Logistic Regression rather than the final SVM, each step was a single run, and the +0.024 test gain was not tested for significance.

Limits common to both datasets. KDEF's controlled studio imagery makes the emotions easy to separate, but that advantage does not carry over. A KDEF-trained SVM reaches a macro-F1 of 0.195 on FER2013, against 0.679 on KDEF, and a simple classifier can tell the two datasets apart from their HOG features with a balanced accuracy of 98.8%. The fuzzy classifier is the most interpretable model, using seven geometric measurements instead of 900 HOG features, and it scores slightly above the SVM on the same test subjects (macro-F1 0.608 against 0.604). Its explanations are still only partly readable, because the linear consequents and overlapping membership functions complicate individual predictions, and its drop from validation to test (0.709 to 0.608) shows limited generalization across people. The live webcam demo is not yet a reliable measure of performance: the informal protocol is not a reliable evaluation metric, fear was very difficult to trigger, and the demo was only tested on macOS, so its behavior on other operating systems is unverified.

References

Goeleven, E., De Raedt, R., Leyman, L., & Verschuere, B. (2008). The Karolinska Directed Emotional Faces: A validation study. *Cognition & Emotion*, *22*(6), 1094–1118. https://doi.org/10.1080/02699930701626582

Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). Challenges in Representation Learning: A Report on Three Machine Learning Contests. In M. Lee, A. Hirose, Z.-G. Hou, R. M. Kil, D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, & G. Weikum (Eds.), *Neural Information Processing* (pp. 117–124, Vol. 8228). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-42051-1_16

A Confusion matrices of the reported models

A.1 FER2013 baseline models (PublicTest; tuned models on PrivateTest)

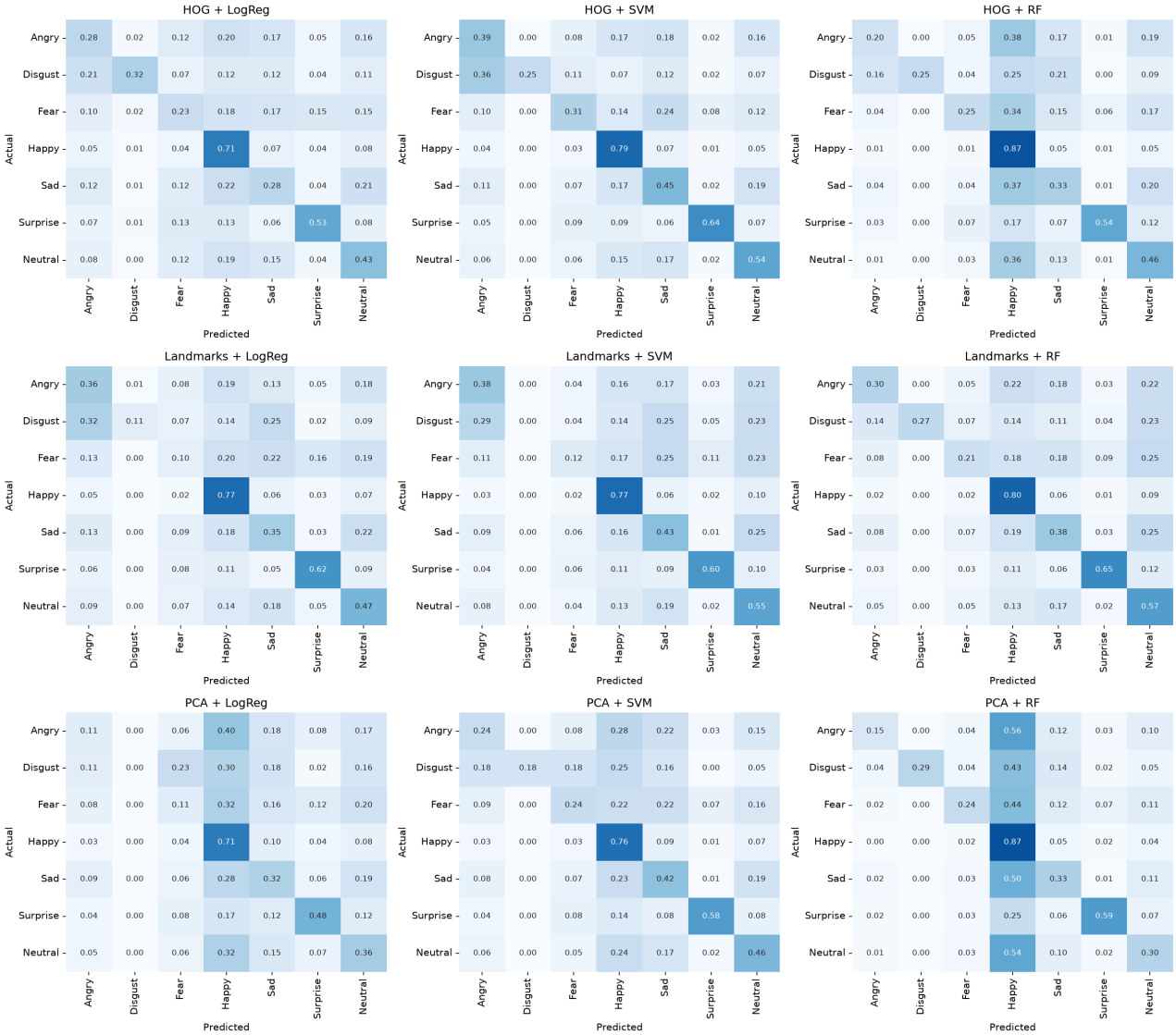


Figure 1: Confusion matrices of all nine FER2013 baseline models on the PublicTest (validation) split. Rows are true labels and columns are predictions; values are row-normalized.

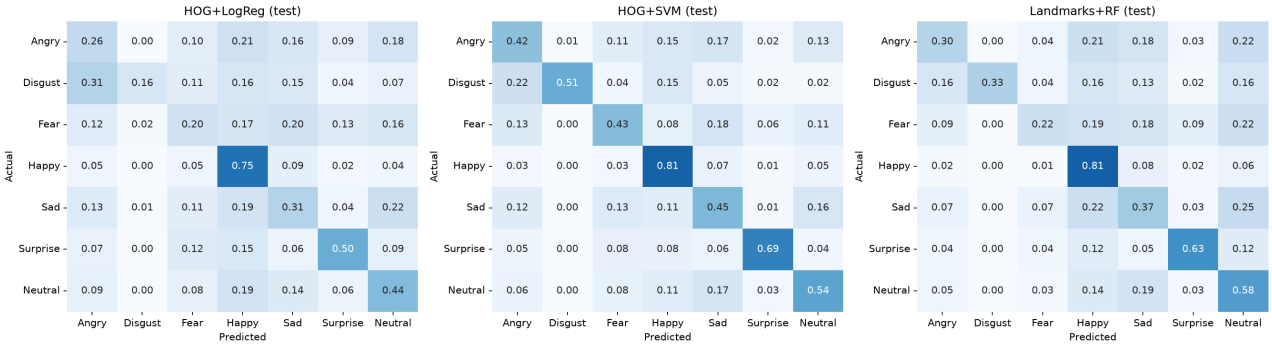


Figure 2: Confusion matrices of the three selected FER2013 models on the PrivateTest split, used for the final evaluation (Table 1). Rows are true labels and columns are predictions; values are row-normalized.

A.2 KDEF study, cross-dataset and creative models

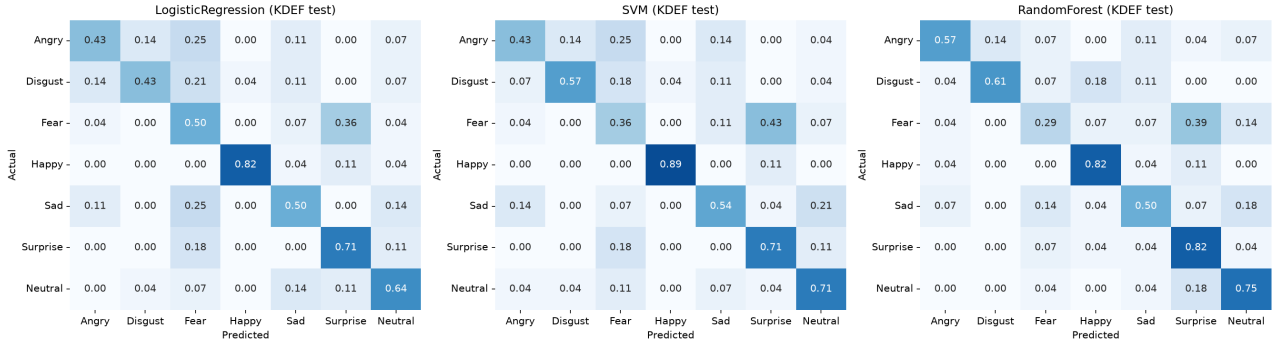


Figure 3: KDEF test subjects (196 images): Logistic Regression, SVM and Random Forest with HOG features, left to right.



Figure 4: KDEF→FER: the Task 2 SVM trained on all KDEF frontal images, evaluated on all FER2013 images. Rows are true labels, row-normalized.

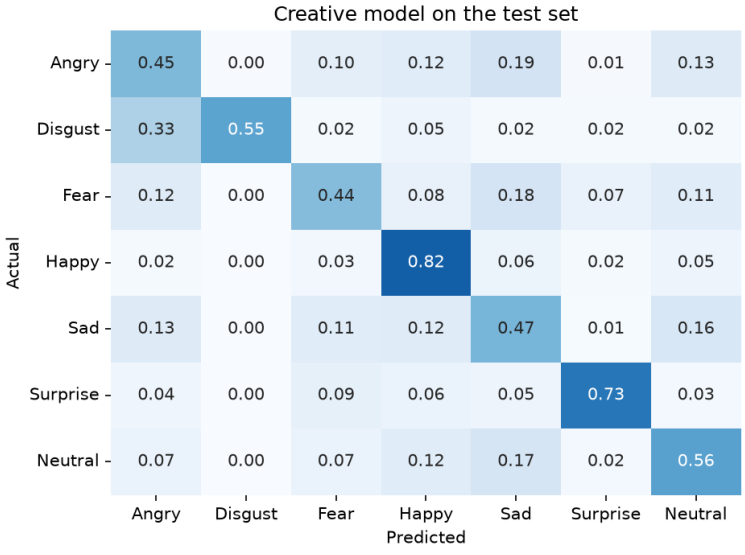


Figure 5: Creative-challenge final model (SVM on HOG + landmarks, cleaned training set), PrivateTest split.

A.3 Fuzzy classifier: held-out KDEF subjects

Table 7: Final fuzzy classifier confusion matrix (counts; 196 test images from 14 unseen subjects). Rows are true labels and columns are predictions.

True / Pred.	Af	An	Di	Ha	Ne	Sa	Su
Afraid	7	1	2	0	2	2	14
Angry	0	14	10	0	2	2	0
Disgust	0	3	23	1	0	1	0
Happy	0	0	0	28	0	0	0
Neutral	0	3	0	0	18	6	1
Sad	0	6	2	0	12	8	0
Surprise	0	0	0	0	1	1	26

Af: afraid; An: angry; Di: disgust; Ha: happy; Ne: neutral; Sa: sad; Su: surprise.

B Supporting tables and implementation details

B.1 Software environments

Environment file	Used for	Versions
`environment-general.yml`	Tasks 1, 2, 3 and 5	Python 3.12.15, scikit-learn 1.9.1, scikit-image 0.26.0, dlib 20.0.1
`environment-fuzzy.yml`	Tasks 4 and 6	Python 3.10.22, NumPy 1.24.4, PyFUME 0.3.4, dlib 19.24.2, scikit-learn 1.7.2

Table 8: Software environments used for each task.

B.2 HOG feature dimensionality

Histogram of Oriented Gradients (HOG) looks at the edges in an image and identifies where the brightness changes and in which direction. To extract the features, the 48×48 grayscale image is cut into a 6×6 grid of cells of 8×8 pixels. In each cell, the change in brightness is measured in 9 different directions $(0°, 20°, \ldots, 160°)$. We then slide a 2×2 block across the grid, which results in 5×5 overlapping block positions:

$$\text{HOG features} = \underbrace{(5 \times 5)}_{\text{block positions}} \times \underbrace{(2 \times 2)}_{\text{cells per block}} \times \underbrace{9}_{\text{directions}} = 900$$

B.3 FER2013 validation results (Task 1)

Table 9: FER2013 PublicTest (validation) results for all nine feature–classifier combinations, with default settings. The selected model is in bold. Metric heatmaps are in Appendix C.2.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1
HOG	Logistic Regression	0.435	0.397	0.395
HOG	**SVM**	**0.542**	**0.482**	**0.505**
HOG	Random Forest	0.482	0.414	0.440
Landmarks	Logistic Regression	0.470	0.397	0.394
Landmarks	SVM	0.500	0.409	0.401
Landmarks	Random Forest	0.514	0.456	0.474
PCA	Logistic Regression	0.382	0.299	0.287
PCA	SVM	0.477	0.410	0.430
PCA	Random Forest	0.454	0.395	0.423

B.4 FER2013 class counts (Task 1)

Table 10: FER2013 class counts per split.

Split	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Train	3,995	436	4,097	7,215	4,830	3,171	4,965
Validation	467	56	496	895	653	415	607

B.5 KDEF grid search space (Task 2)

Classifier	Hyperparameter	Values searched
Logistic Regression	Scaler	none, Standard, Robust, MinMax, **PowerTransformer**
	C	**0.01**, 0.1, 1, 10
	20 configurations	
SVM	Scaler	Standard, Robust, **PowerTransformer**
	Kernel	linear, **RBF**, polynomial
	C	0.01, 0.1, 1, **10**
	γ (RBF only)	0.01, 0.001, **0.0001**, auto
	Degree (polynomial only)	2, 3, 4
	96 configurations	
Random Forest	Number of trees	100, **200**
	Maximum depth	3, 5, 8, **10**, 15, 20, unlimited
	Minimum samples per leaf	1, **2**, 5, 10, 20, 50
	84 configurations	

Table 11: Hyperparameter tuning grid for KDEF: Selected values (highest macro-F1) are in bold

B.6 Per-emotion F1 on KDEF and FER2013 (Task 2)

	KDEF			FER2013
Emotion	Logistic Regression	SVM	Random Forest	SVM
Angry	0.50	0.50	0.65	0.44
Disgust	0.53	0.65	0.69	0.62
Fear	0.41	0.33	0.34	0.46
Happy	**0.88**	**0.93**	**0.77**	**0.75**
Sad	0.51	0.55	0.53	0.43
Surprise	0.63	0.62	0.63	0.73
Neutral	0.61	0.66	0.69	0.55

Table 12: F1 score per emotion - classifier comparison

B.7 Per-emotion F1 of the cross-dataset model (Task 3)

Happy has the best numbers probably because a happy face always has a smile, and the edges around the lips are similar in both datasets.

	Happy	Surprise	Sad	Angry	Fear	Disgust	Neutral
F1 on KDEF (cross-validation)	0.89	0.71	0.50	0.66	0.46	0.69	0.66
F1 on FER2013	0.38	0.31	0.24	0.19	0.18	0.04	0.02

Table 13: F1 per emotion of the KDEF-trained SVM on KDEF and on FER2013.

B.8 HOG features that transfer best and worst (Task 3)

Transfer best				Transfer worst			
Row	Column	Direction	Correlation	Row	Column	Direction	Correlation
5	3	20	0.95	1	5	160	-0.88
2	3	80	0.95	5	5	0	-0.86
3	4	80	0.94	1	2	80	-0.84
4	3	40	0.93	5	0	160	-0.80
4	3	60	0.93	1	5	160	-0.78

Table 14: The five HOG features that transfer best and worst from KDEF to FER2013. Row and column give the cell in the 6×6 grid (row 0 is the top of the image, column 0 the left), and direction is the edge angle in degrees. The same cell can appear twice because it belongs to up to four overlapping blocks.

B.9 Example fuzzy rule (Task 4)

Full RULE1 of the happiness model, as generated by PyFUME (`outputs/fuzzy/pyfume_work/final/3_clusters/happy/Sim`

> *Happy, Rule 1:* IF mouth_width IS cluster1 AND mouth_opening IS cluster1 AND left_eye_opening IS cluster1 AND right_eye_opening IS cluster1 AND eyebrow_height IS cluster1 AND smile_ratio IS cluster1 AND mouth_corner_asymmetry IS cluster1, THEN OUTPUT $= -0.753203 + 1.537483\,\text{mouth_width} + 0.78748\,\text{mouth_opening} - 0.914991\,\text{left_eye_opening} + 0.889586\,\text{right_eye_opening} + 0.18434\,\text{eyebrow_height} + 0.408057\,\text{smile_ratio} - 0.079164\,\text{mouth_corner_asymmetry}$.

Rules 2 and 3 of the same model use the same structure with cluster2 and cluster3 and their own linear consequents (`fun2` and `fun3` in the same file).

B.10 Hyperparameter tuning details (Task 1)

The SVM grid covered $C \in \{1, 5, 10\}$ and $\gamma \in \{0.5, 1, 2\} \times 1/900$, tuned on a 10,000-image subsample of the training set; the Logistic Regression grid covered $C \in \{0.01, 0.1, 1, 10\}$; and the Random Forest grid covered maximum depth (unlimited or 25) and minimum leaf size (1, 3 or 5). The best settings were refit on the full training split and scored on validation: the SVM reached 0.562 macro-F1 ($C = 10$, $\gamma = 1/900$), against 0.505 by default; Logistic Regression ($C = 0.01$) reached 0.411; and the landmark Random Forest (maximum depth 25, minimum leaf 1) reached 0.474.

B.11 Feature and classifier discussion (Task 1)

We attribute the HOG result to its capture of local texture and edge information that distinguishes expressions (open mouth, raised or furrowed brows, wrinkles around the eyes and nose), with some invariance to illumination. Landmarks discard texture, and at 48×48 pixels the dlib detector works with very low resolution, so the landmark positions are noisy. Landmarks are also computed with the full image as the face box, which assumes FER crops are well aligned; this is not always true (see the error analysis in Appendix C). PCA is a holistic linear projection of raw pixels, so it is sensitive to misalignment, pose and lighting.

The most frequent errors of the validation SVM are Fear to Sad (24% of all Fear), Sad to Neutral (19%), Angry to Sad (18%), Sad to Happy (17%) and Neutral to Sad (17%).

A plausible but untested reason for the landmark result is that the forest can pick out a few informative coordinates from the 136 landmark values, while the RBF kernel, with standardized inputs and equal weight on all coordinates, is diluted by the uninformative ones.

For the demo, prediction time is 0.16 ms per image for Logistic Regression, 32 ms for the SVM, and 27 ms plus 11 ms of feature extraction for the Random Forest.

B.12 KDEF image consistency (Task 2)

The HOG features of two happy faces look very similar, and the classifiers need few examples to learn each emotion. As mentioned before, the images in the KDEF dataset are consistent and is balanced. Each facial attribute is present in a similar pixel vicinity across all images of an emotion. This makes the models learn better, unlike the FER2013 where there is no consistency in the subjects present, alignment and facial features/attributes of the face etc.

B.13 Rule activation (Task 4)

The strongest rules for fear and sadness showed weaker separation, and overlapping membership functions limit their linguistic interpretation.

B.14 Dataset classifier (Task 3)

To check whether the datasets also differ in the HOG features the model uses, we trained a simple classifier to predict which dataset an image came from. We used all 980 KDEF images and a random sample of 980 FER2013 images (140 per emotion randomly chosen), so both sets had the same size and the same emotion mix. The classifier reached a balanced accuracy of 98.8% (5-fold cross-validation).

B.15 HOG spatial mechanism (Task 3)

HOG splits the image into a fixed grid of cells and records the direction of the edges in each cell, so every feature is tied to a specific location in the image. In KDEF, the eyes, eyebrows and mouth always fall into the same cells, and the model learns which cells matter for each emotion. In FER2013, faces are often shifted, tilted or cropped differently, as the mean faces in Figure 11 show, so the same facial features land in different cells. Even when the expression is identical, the resulting gradient pattern can look very different from anything the model saw during training.

B.16 Subject split and fuzzy classifier details (Task 4)

The predefined subject-independent split contained 56 training subjects (784 images) and 14 held-out test subjects (196 images). Within the training partition, 44 subjects (616 images) were used for fitting and 12 (168 images) for validation. MinMax scaling was fitted only on the corresponding training partition.

B.17 Selected KDEF model settings (Task 2)

The selected settings are a PowerTransformer scaler with $C = 0.01$ for Logistic Regression, a PowerTransformer scaler with an RBF kernel ($C = 10$, $\gamma = 0.0001$) for the SVM, and 200 trees with maximum depth 10 and at least 2 samples per leaf for the Random Forest.

B.18 PyFUME parameters (Task 4)

The parameters altered from the PyFUME defaults are $setnes_threshold = 1.0$, $merge_threshold = 1.0$ and $data_split_method =$ *"no_split"*. Together they stop PyFUME from removing all fuzzy sets of a variable during rule simplification and from withholding part of the fitting data.

B.19 Demo run and detection details (Task 6)

We added a **README** file in the root folder of the submitted files explaining how to run the demo in the terminal. The same results can be obtained by running the notebook `notebooks/06_task6_webcam_demo.ipynb`.

Detection was done differently from training, since training upsampled once on the original *KDEF* photos. The demo uses no upsampling on the downscaled frame.

B.20 Live demo evaluation protocol (Task 6)

- Refer to an emotion from a **KDEF** reference
- Mimic the emotion and start the simulation
- Run the simulation for a while without changing the emotion displayed
- Compute the results

B.21 Duplicate analysis (Task 1)

Of the 3,301 non-duplicated test images, the SVM reaches 54.2% accuracy and 0.505 macro-F1; the 288 duplicated test images (8.0%) reach 95.5% accuracy and 0.943 macro-F1, and 96.9% of the duplicates carry the same label as the training copy. The reported test score of 0.568 macro-F1 is therefore optimistic, and the non-duplicated

score of 0.505 is the fairer estimate. On validation, 280 of 3,589 images (7.8%) also appear in training, and removing them lowers the SVM's macro-F1 from 0.505 to 0.438. The drop is largest for the Random Forests (0.097 to 0.110), which memorize training samples, and smallest for the landmark models (0.008 to 0.009).

B.22 Fuzzy classifier accuracy comparison (Task 4)

The fuzzy model reaches 63.27% accuracy on the held-out subjects, the highest of the three classifiers in Table 2 (SVM 60.20%, Random Forest 62.24%, Logistic Regression 57.65%). These gaps correspond to a few images out of 196. The validation drop of the fuzzy model (0.709 to 0.608) is larger than that of the SVM (0.679 to 0.604), but the two are not directly comparable: the SVM value is 5-fold cross-validation on the training subjects, whereas the fuzzy value comes from a single 12-subject validation split.

B.23 KDEF collection protocol (Task 2)

The Karolinska Directed Emotional Faces - KDEF was a dataset that was generated in 1998. It has a total of 4900 pictures, of 70 individuals displaying 7 different emotions. The subjects for this dataset was chosen with a specific selection criteria some of which include having no beard, mustache, earrings etc. All subjects were given special gray shirts and all the photos were taken at a distance of 3m from the camera. Each subject was taken photos in 2 sessions, 5 poses for each of the 7 emotions. The pose that is of most interest to us is *straight*.

Unlike the FER2013 dataset (which are 48×48 dimensions), the KDEF dataset has images of dimensions 562×762 and RGB images. To compare the 2 datasets, the KDEF images need to be converted to the FER2013 format: a Haar cascade detects the face, the box is enlarged by 20% so the whole face and some context are kept, and the crop is converted to grayscale, resized to 48×48 and scaled to $[0, 1]$.

C Supporting figures and qualitative examples

C.1 FER2013 class distribution and error analysis (Task 1)

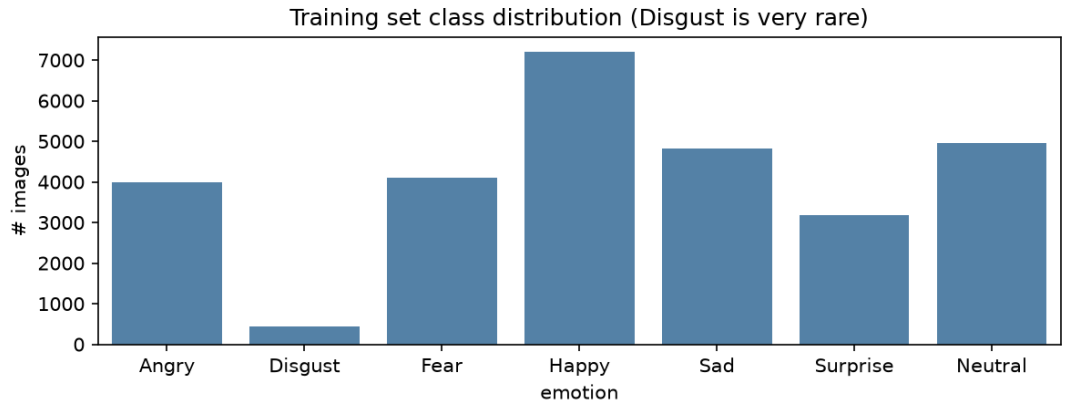


Figure 6: FER2013 training set: class distribution.

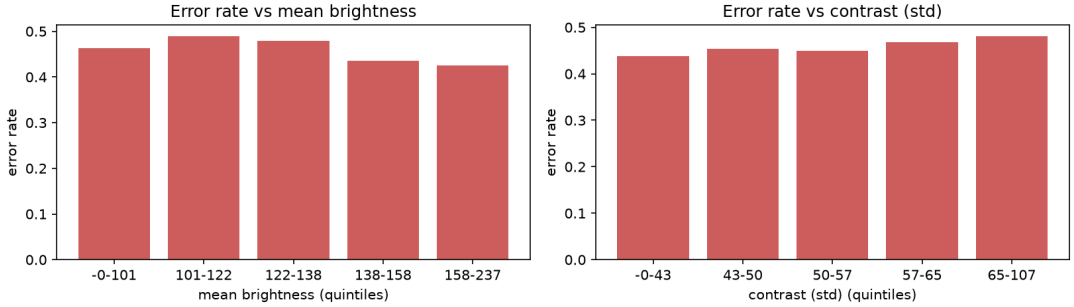


Figure 7: Error rate against lighting (HOG + SVM, validation split).

Illumination and contrast have a small effect on error. We grouped validation images into quintiles of mean brightness and of contrast (pixel standard deviation) and computed the error rate of the SVM in each group. The error rate is about 0.46 to 0.49 for the three darkest quintiles and about 0.42 to 0.44 for the brightest two, and it rises slightly from about 0.44 to 0.48 as contrast increases (Figure 7). The effect is around five percentage points, so illumination matters but is not the main source of error.

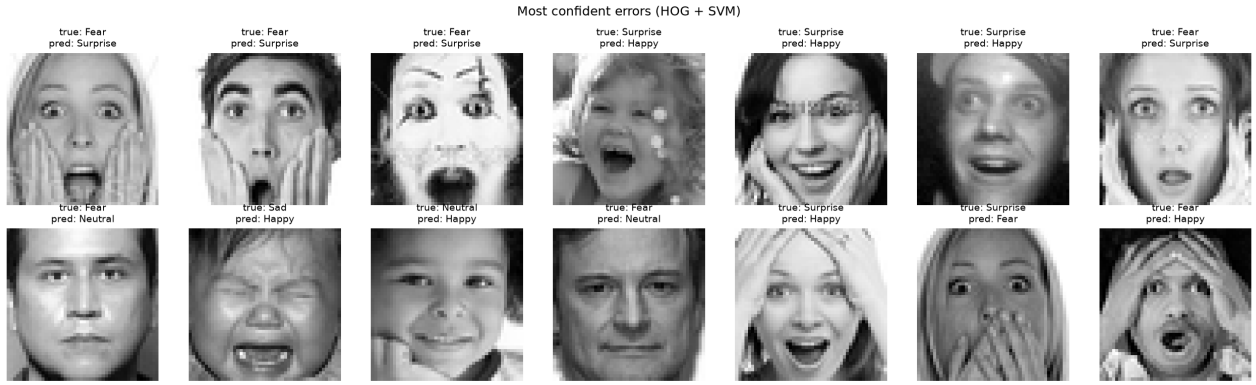


Figure 8: Random sample of misclassified FER2013 validation images.

Figure 8 shows recurring problems that are properties of the dataset rather than of the model: *ambiguous or questionable labels* (a screaming infant labelled Angry and predicted Happy; several wide-eyed, open-mouthed faces labelled Neutral or Fear), *occlusion and context* (hands over the face, hats, two faces in the crop), *non-frontal pose and head tilt*, which break the aligned-face assumption of all three feature types, and *population and artefacts* (infants and elderly faces, overlaid text or watermarks). The gallery is a random sample of errors, so it is qualitative evidence, not a count of mislabelled images.

C.2 Validation metric heatmaps (FER2013)

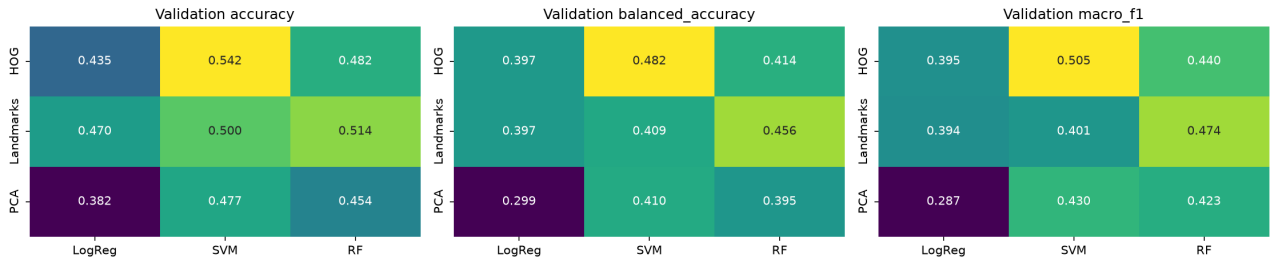


Figure 9: Metric heatmaps of each feature–classifier combination on the FER2013 PublicTest (validation) split.

C.3 KDEF split and mean faces (Task 2)

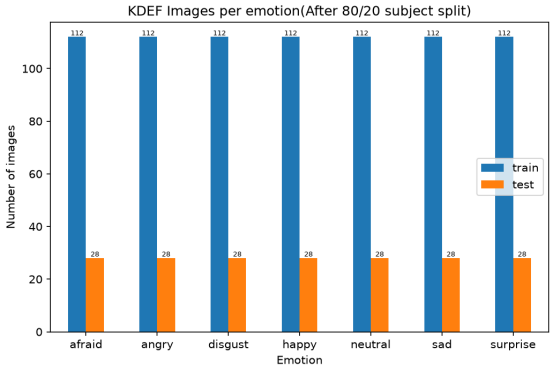


Figure 10: Number of KDEF images per emotion in the training (56 subjects) and test (14 subjects) split.

To understand how these alignment of these facial features could be of concern for the model, we generated the mean face for each emotion in both datasets. This was done to understand how well positioned each facial feature is in all images across the dataset. The more blurry the images are, the more inconsistent the images are in the dataset for each emotion.

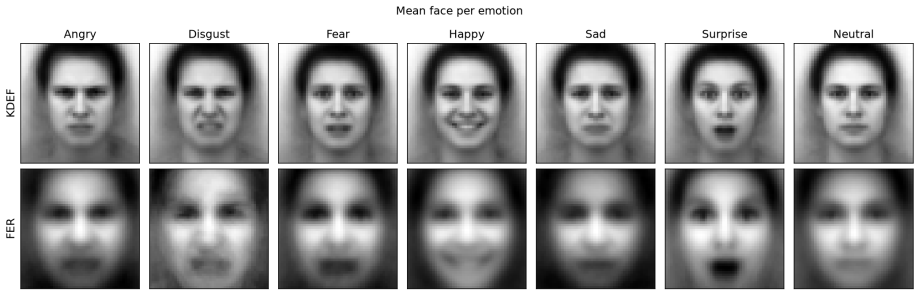


Figure 11: Mean face per emotion of KDEF and FER2013 training images

C.4 Domain shift and transfer examples (Task 3)

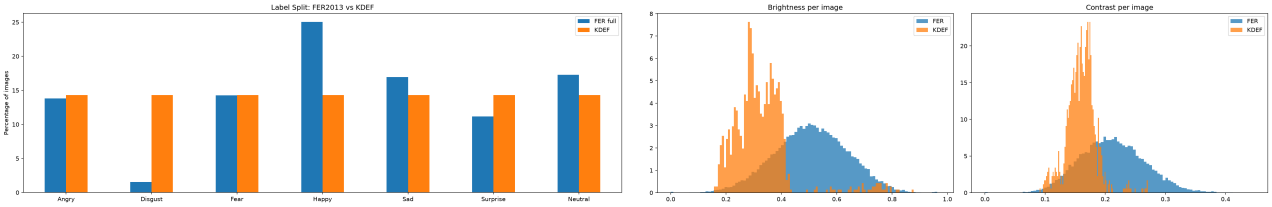


Figure 12: Left: share of each emotion in FER2013 and KDEF. Right: brightness and contrast of the images in both datasets.

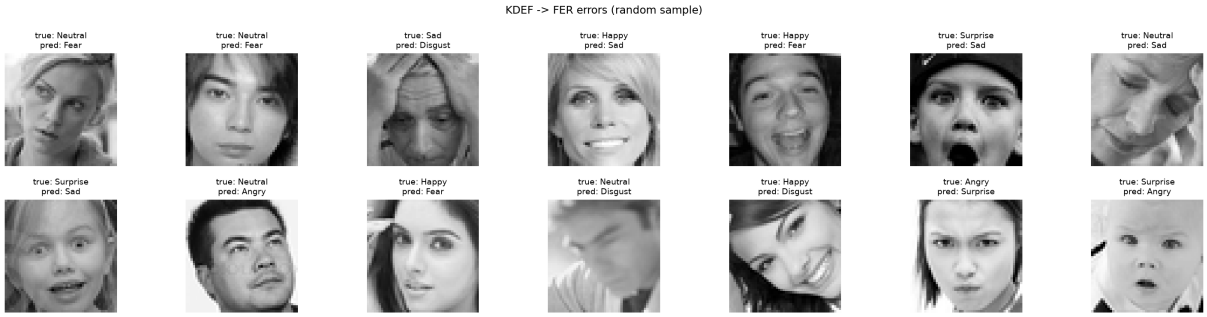


Figure 13: Random FER2013 images that the KDEF-trained model gets wrong, with the true and predicted emotion.

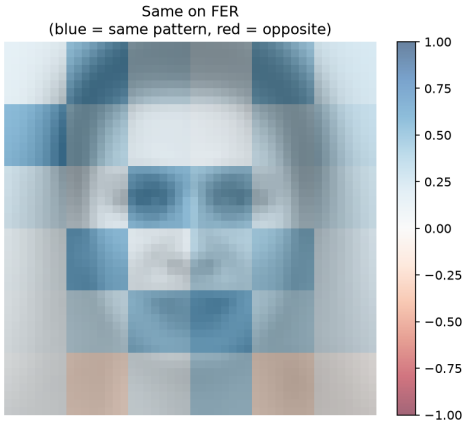


Figure 14: How well the HOG features in each region of the face transfer from KDEF to FER2013 (blue: same pattern in both datasets; red: opposite pattern).

C.5 Live demo qualitative comparison (Task 6)

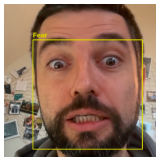
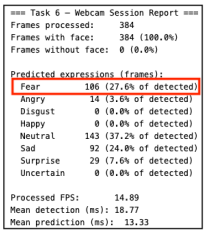
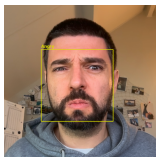
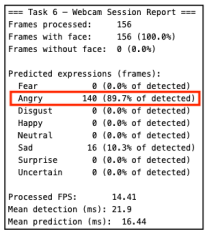
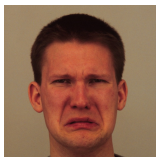
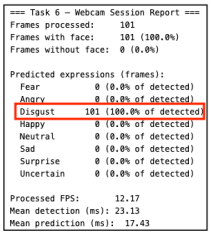
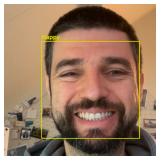
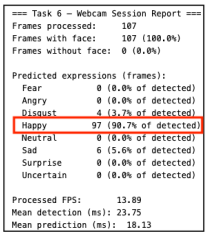
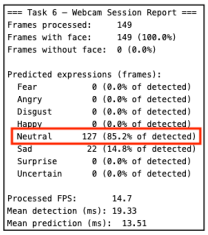
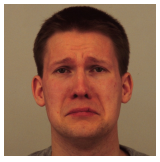
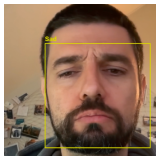
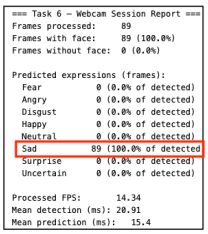
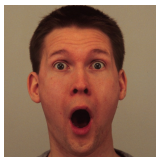
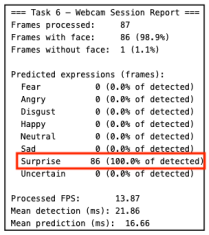
Posed expression	KDEF reference	Webcam capture	Prediction
Fear			
Angry			
Disgust			
Happy			
Neutral			
Sad			
Surprise			

Table 15: Qualitative comparison of posed KDEF expressions, corresponding webcam expressions, and fuzzy classifier predictions. KDEF subject M11, acquisition A, frontal pose. Webcam participant was the group member Henrique da Silva Couto. This illustrative experiment does not establish classification accuracy.